

Worksheet 2: Solutions

The Qubit and Quantum Mechanics Fundamentals

TA: Haripriya | haripriya.pettugani@unibas.ch

1.

- (a) $p(0) = |\alpha|^2 = 3/4$ and $p(1) = |\beta|^2 = 1/4$, so $|\alpha| = \sqrt{3}/2$ and $|\beta| = 1/2$. The moduli say nothing about the phases. A normalised state has four real numbers, normalisation removes one and the global phase another, leaving two; fixing $|\alpha|$ uses one of those, so **one real parameter** (the relative phase) remains free.

- (b) With $\alpha = \sqrt{3}/2$ and $\beta = \frac{1}{2}e^{i\varphi}$,

$$p(+) = \frac{1}{2}|\alpha + \beta|^2 = \frac{1}{2}\left(\frac{3}{4} + \frac{1}{4} + 2 \cdot \frac{\sqrt{3}}{2} \cdot \frac{1}{2} \cos \varphi\right) = \frac{1}{2}\left(1 + \frac{\sqrt{3}}{2} \cos \varphi\right).$$

Setting this to $1/2$ gives $\cos \varphi = 0$, so $\varphi = \pi/2$ or $\varphi = -\pi/2$, i.e. $\beta = \pm i/2$.

- (c) $p(+i) = \frac{1}{2}|\alpha - i\beta|^2$. For $\beta = +i/2$, $\alpha - i\beta = \frac{\sqrt{3}}{2} + \frac{1}{2}$ and

$$p(+i) = \frac{1}{2}\left(\frac{\sqrt{3}+1}{2}\right)^2 = \frac{2+\sqrt{3}}{4} = 0.933, \quad p(-i) = 0.067.$$

For $\beta = -i/2$ the two values are exchanged. The two candidates are therefore indistinguishable under Z and under X but maximally different under Y , so the third experiment must be a Y measurement.

2.

- (a) Taking the plate to act on the $|1\rangle$ arm: $|0\rangle \xrightarrow{H} \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \xrightarrow{\text{phase}} \frac{1}{\sqrt{2}}(|0\rangle + e^{i\varphi}|1\rangle)$, and applying H again,

$$\frac{1}{2}[(|0\rangle + |1\rangle) + e^{i\varphi}(|0\rangle - |1\rangle)] = \frac{1}{2}[(1 + e^{i\varphi})|0\rangle + (1 - e^{i\varphi})|1\rangle].$$

Then $p(0) = \frac{1}{4}|1 + e^{i\varphi}|^2 = \frac{1}{4}(2 + 2\cos \varphi) = \cos^2(\varphi/2)$, and $p(1) = \sin^2(\varphi/2)$.

Had the plate been placed in the $|0\rangle$ arm instead, the state before the second H would be $\frac{1}{\sqrt{2}}(e^{i\varphi}|0\rangle + |1\rangle)$, which is the above multiplied by the global phase $e^{i\varphi}$ with $\varphi \rightarrow -\varphi$; the two probabilities come out interchanged, $p(0) = \sin^2(\varphi/2)$. Either convention is correct provided it is stated, and the physics — full fringe contrast, period 2π — is the same.

- (b) With the plate in the $|1\rangle$ arm: $\varphi = 0$ sends every photon to the $|0\rangle$ detector, $\varphi = \pi/2$ splits the beam evenly, and $\varphi = \pi$ sends every photon to the $|1\rangle$ detector. With the other convention the first and last are exchanged.
- (c) With a which-path detector the two arms are no longer alternatives for one photon but two separate experiments, each occurring half the time. A photon known to be in one arm meets the second beam splitter alone and leaves with $p(0) = p(1) = 1/2$; the same holds for the other arm. Averaging gives $p(0) = p(1) = 1/2$ for every φ .

What is lost is the cross term: in (a) the two amplitudes were added before squaring, and φ entered only through that interference. Once the path is recorded, probabilities are added instead of amplitudes, and φ drops out.

3.

- (a) The atoms arrive in $|0\rangle$, so the survival probability is $|\langle n_+ | 0 \rangle|^2 = \cos^2(\theta/2)$.
- (b) They leave in $|n_+\rangle$, so the third magnet gives $p(1) = |\langle 1 | n_+ \rangle|^2 = \sin^2(\theta/2)$.

- (c) The product is

$$\cos^2(\theta/2) \sin^2(\theta/2) = \frac{1}{4} \sin^2 \theta,$$

using $\sin \theta = 2 \sin(\theta/2) \cos(\theta/2)$. This is maximised at $\theta = 90^\circ$, where it equals $1/4$ — the case done in the lecture. Values: $1/16$ at 30° , $3/16$ at 60° , $1/4$ at 90° , $3/16$ at 120° .

- (d) At $\theta = 0$ the middle magnet is parallel to the first: every atom survives it and none reaches the $|1\rangle$ output. At $\theta = 180^\circ$ the middle magnet is anti-parallel, so its $+$ output is empty and nothing gets through at all. In both cases the middle magnet asks the same question as the first, or its exact negation, and so tells us nothing new; a classical filter would behave identically. The effect is largest where the two questions are maximally incompatible.

4.

- (a) The first maximum is at $\Omega t = \pi$, so $\Omega = \pi/25 \text{ ns} = 1.257 \times 10^8 \text{ s}^{-1}$ and $\Omega/2\pi = \mathbf{20.0 \text{ MHz}}$. The equator is reached at half that pulse area, so $t = \mathbf{12.5 \text{ ns}}$.
- (b) $\frac{\Omega^2}{\Omega^2 + \Delta^2} = 0.64 \Rightarrow \frac{\Delta^2}{\Omega^2} = \frac{1}{0.64} - 1 = 0.5625 \Rightarrow \Delta/\Omega = \mathbf{0.75}$.
- (c) $\sqrt{\Omega^2 + \Delta^2} = \Omega\sqrt{1.5625} = 1.25\Omega$, so the observed frequency is $1.25 \times 20.0 = \mathbf{25.0 \text{ MHz}}$. The tilt is $\arctan(0.75) = \mathbf{36.9^\circ}$.
- (d) The on-resonance rate Ω is already known from the first qubit, since both are driven at the same amplitude. Measuring the oscillation frequency of the second qubit gives $\sqrt{\Omega^2 + \Delta^2}$ directly, and Δ follows: $\Delta = \sqrt{(1.25\Omega)^2 - \Omega^2} = 0.75\Omega$. A frequency is usually the easier quantity to measure accurately, since it does not depend on the readout being perfectly calibrated, whereas the contrast does.

5.

- (a) $C|+\rangle|0\rangle = \frac{1}{\sqrt{2}}(C|0\rangle|0\rangle + C|1\rangle|0\rangle) = \frac{1}{\sqrt{2}}(|0\rangle|0\rangle + |1\rangle|1\rangle)$.
- (b) $|+\rangle|+\rangle = \frac{1}{2}(|0\rangle|0\rangle + |0\rangle|1\rangle + |1\rangle|0\rangle + |1\rangle|1\rangle)$, which has four terms where (a) has two. The states are different, so C fails to copy $|+\rangle$. Since C was an arbitrary gate satisfying the two basis conditions, no gate copies an unknown state.
- (c) ★ Suppose C copies both $|a\rangle$ and $|b\rangle$. Unitary evolution preserves inner products, so comparing the states before and after gives $\langle a|b\rangle = \langle a|b\rangle^2$, whence $\langle a|b\rangle = 0$ or 1 . A copier can therefore exist only for states that are either identical or orthogonal, which is to say only for states some single measurement already distinguishes perfectly. That is the classical case, and it is why error correction cannot store duplicates: the states it must protect are superpositions, which are not orthogonal to the basis states used to make copies. It works with parity measurements instead, which reveal whether two qubits agree without revealing what either one is.

6. ★

- (a) Scalable, well-characterised qubits; initialisation to a known state; coherence long compared with gate times; a universal gate set; qubit-specific measurement.
- (b) Any two, for example: superconducting circuits are comfortable on gate speed and fabrication but strained on coherence and on fixed nearest-neighbour connectivity; trapped ions are comfortable on coherence, fidelity and connectivity but strained on gate speed and on scaling a single chain.
- (c) Error correction closes the gap, by encoding one logical qubit in many physical ones. The threshold is the physical error rate below which enlarging the code suppresses the logical error rate; above it, adding qubits makes matters worse rather than better.

7.

- (a) Light emerging from any filter is polarised along that filter's axis. The next filter is rotated by $\vartheta = 90^\circ/N = \pi/2N$, so it transmits a fraction $\cos^2(\pi/2N)$ and re-polarises the survivors along its own axis. The situation after each filter is therefore identical to the situation before it, rotated. There are N such steps, and the probabilities multiply:

$$T(N) = [\cos^2(\pi/2N)]^N = \cos^{2N}(\pi/2N).$$

- (b) $T(2) = \cos^4 45^\circ = \frac{1}{4}$, which is the three-filter experiment: a quarter of the light gets through where none did before. $T(10) = \cos^{20} 9^\circ = 0.78$ and $T(100) = \cos^{200} 0.9^\circ = 0.976$.
- (c) For large N , $\cos(\pi/2N) \approx 1 - \frac{1}{2}(\pi/2N)^2$, so

$$\ln T = 2N \ln \cos(\pi/2N) \approx 2N \left(-\frac{1}{2}\right) \left(\frac{\pi}{2N}\right)^2 = -\frac{\pi^2}{4N},$$

giving $T(N) \approx e^{-\pi^2/4N}$ and $T \rightarrow 1$ as $N \rightarrow \infty$. (At $N = 100$ the approximation gives 0.9756, matching the exact value to four figures.)

- (d) Each filter does measure, and each measurement does destroy the information established by the previous one. The point is how much is destroyed. A measurement in a basis tilted by ϑ costs a probability of order ϑ^2 , while the rotation it achieves is of order ϑ . Taking N steps accumulates a rotation of order $N\vartheta$ but a loss of order $N\vartheta^2$, and with $\vartheta = \pi/2N$ the loss is $\pi^2/4N \rightarrow 0$. Measurement is not gentle in absolute terms, but it is gentle at second order, and that is enough for many small measurements to steer a state where one large measurement would have destroyed it. The same quadratic smallness underlies the quantum Zeno effect, in which repeated rapid measurement freezes a state instead of randomising it.